query might return something akin to what KaZaA displays—Our Computer: 70,65,345 users, 65,43,58,723 processes; Free resources: 44,58,92,87,345 GB; 8,69,307 GHz!

Who Will Govern UC?

A valid question, but one that still has no clear answers. In order for UC to become a reality, governments will need to get involved. The infrastructure required is tremendous, and more than any individual company can finance. Of course, software also has to be written to support and share the hardware.

Another possibility is that, as with SETI@home, groups may form to freely share resources and form private grids. In this scenario, everyone buys hardware and software and then joins their favourite UC group. Perhaps private groups will charge a nominal fee to become a member and review each member's contribution, based on which the annual fee would increase or decrease.

Another possibility is to leave UC to care for itself, as happened with the Internet. Just a few governing bodies here and there to prevent total mayhem, but we'd still have the gigantic processing power of every computer online!

What About Privacy And Security?

The greatest fear of using UC will stem from privacy and security. Every computer out there, whether at work or at home, contains some information that the owner considers private. People fear that joining a UC grid is akin to publishing all your hard drive's content on the Internet—and with write access!

Perhaps that's why governments need to be involved. We need the accountability that is sorely lacking with the Internet. The threat of spyware, adware, viruses and worms from the Internet is already invading our privacy—we don't need less privacy, we need more. Software and hardware vendors will claim to design products that offer you privacy, but they promise that already, and we already know that everything tech is fallible.

What About Other Devices?

It's not only computers that will benefit. With pervasive computing and convergence becoming reality, soon everyone will know how to use gadgets, and will want to only work with gadgets. We will forget about, say, standard refrigerators. We will want a single input device, like the computer, and expect everything else to know what our computer does.

Let's say you plan a two-week vacation; you tell your computer this; the computer informs all your gadgets. The fridge informs you that you better remove the ripe tomatoes and the Macaroni Salad, lest you want to come back to a messy and smelly fridge. The washing machine reads the tags on your clothing and tells you that you need to do a quick wash if you plan to take your best clothes. The car informs you that you haven't had it serviced this month, and you definitely need to do so before your long drive. Your mobile automatically selects and chooses the right roaming plan. You give the go ahead, fall asleep, your body monitoring-watch and alarm clock communicate and find the ideal time to awaken you—they consider how much rest you have got according to your body's readings, and also how much time you normally take to get ready at this time of the day. You wake up, your hot bath is already waiting for you...

This scenario seems out of place when talking about Utility Computing, but is a necessary example to inform you that everything will have an embedded chip soon, and everything will be connected. Universal resources are the key; and just as they are not hardware or software specific, the are not device or task specific either.

Communications Central

Everything will be able to communicate, and share resources

Info-graphics Binesh Sreedharan

Is Any Of This Actually Possible?

All of the above was theory of course, but theory based on facts. (*See boxes 'Sun's Grid' and 'The Cell'*) The fact is, companies have just started releasing results of years of research. The Cell, for example, was developed by IBM, Toshiba and Sony. It will be the first processor that can share resources with similar processors.

Sun's UC plans also prove that there is a market for on-demand computing, and that the bigwigs are sitting up and taking notice. UC needs the likes of Sun, IBM, Sony, and others, to take interest in this promising field. Not only will this increase competition, it will speed up the creation of solutions to the problems faced by UC.

As technology becomes more pervasive than ever, your PC might soon wield more power than today's supercomputers. And all this will come at the price of your household utility bills! ⊠

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Sun's Grid

Sun Microsystems Inc. is ready to unveil its Sun Grid. It plans to open six grid centres around the world.

Customers can access the grid and are charged a mere $1 per hour per CPU. Out of the six data centres, three are in the US, two in the UK, and one in Canada.

The servers consist of both Sun's SPARC and AMD's Opteron processors, all running Solaris 10. This is the first step to managed UC being offered around the world. Sun plans to have a global grid, but with regional services.